Chapter 20 — Credential & Token Storage (Android Keystore)

> **Part III — Authentication & Security** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 4 (DataStore — where *non-secret* config lives), Ch 19 (token refresh — what we're storing), Ch 21 (biometrics — auth-bound keys). This chapter **defines** the `TokenStore`/secret-storage contract that Ch 19, Ch 22 (logout), and Ch 42 (multi-account) depend on.
>
> **Reference surface in StreamCart:** storing the refresh token, a Wallet PIN hash, and an encryption key for the offline media/DB — safely, on a device the user may have rooted.

20.1 Problem Definition and Real-World Context

Apps must persist secrets — refresh tokens (Ch 19), encryption keys, PINs — on a device that is **not fully trusted**: it may be rooted, backed up to the cloud, shared, or stolen. Storing secrets naively is one of the most common, highest-severity mobile vulnerabilities:

- **Plaintext in SharedPreferences/DataStore/SQLite.** On a rooted device (or via `adb backup`, or a backup to the cloud) these files are trivially readable. A refresh token in plaintext = full account takeover.
- **Hardcoded keys in the APK.** Decompiling an APK exposes any key baked into code/resources; "encryption" with an embedded key is not encryption.
- **Keys recoverable from the app process.** If the encryption key lives in memory as a plain byte array that's persisted, extracting it defeats the scheme.
- **Secrets surviving logout / leaking across accounts.** Not clearing on logout (Ch 22) leaves a valid token behind; not scoping per account (Ch 42) leaks.

The **Android Keystore** solves the core problem: it generates and stores cryptographic keys in a **hardware-backed, isolated** container (TEE/StrongBox) where the **key material never enters the app process** — the app can *use* the key to encrypt/decrypt but cannot *extract* it. The pattern is **envelope encryption**: a Keystore key encrypts the secret; the ciphertext goes in ordinary storage. Optionally, keys are **bound to user authentication** (biometric/PIN, Ch 21). Understanding exactly **what the Keystore does and does not protect** is the architectural crux.

Three practical use cases

1. **Refresh token (StreamCart Account).** Encrypt with a Keystore-backed key; store ciphertext in DataStore; decrypt on demand for refresh (Ch 19); clear on logout.
2. **Wallet secret (Fintech).** A biometric-bound Keystore key gates decrypting a payment token — the secret is unusable without a fresh biometric (Ch 21).
3. **Offline DB/media key (OTT).** A Keystore key encrypts the SQLCipher/media key (Ch 24) so downloaded content can't be read off a stolen device.

Why this is architectural

Secret storage defines the app's **trust boundary** (what the hardware protects vs what the OS/app can), the **threat model** (rooted device, backup, theft), and the **key lifecycle** (generation, auth-binding, invalidation, rotation, clearing) — properties that determine whether a compromised device means a compromised account.

20.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	Store secrets (tokens, keys, PIN hashes) encrypted at rest.
F2	Encryption keys live in the Android Keystore (never extractable).
F3	Prefer hardware-backed/StrongBox keys where available.
F4	Optionally bind key use to user authentication (biometric/PIN, Ch 21).
F5	Exclude secrets from backups.
F6	Clear secrets on logout (Ch 22) and scope per account (Ch 42).
F7	Handle key invalidation (new biometric enrolled, lock removed, key lost).

Non-functional requirements

Attribute	Policy
Security	Keys non-extractable; hardware-backed; auth-bound for sensitive.
Reliability	Graceful handling of key invalidation / `KeyPermanentlyInvalidatedException`.
Compatibility	Works across API levels and devices (StrongBox optional).
Performance	Crypto ops cheap; avoid per-request key gen.
Maintainability	A single secret-storage abstraction (`SecureStore`).

Scope

- **In scope:** Android Keystore key generation (AES/GCM), envelope encryption of secrets, StrongBox, auth-bound keys (intro; depth in Ch 21), backup exclusion, invalidation handling, clearing/rotation, the `SecureStore`/`TokenStore` contract.
- **Out of scope:** biometric prompt flow (Ch 21), DB/file encryption specifics (Ch 24), network security (Ch 23), the refresh logic (Ch 19), root detection (Ch 26).

Assumptions & constraints

- `minSdk 24`; `AndroidKeyStore` provider; AES-256-GCM. Note: `androidx.security:security-crypto` (EncryptedSharedPreferences) is **deprecated** — this chapter uses **Keystore directly** (and mentions Tink).

20.3 System Design Fundamentals

- **Android Keystore.** A system-provided `KeyStore` (`"AndroidKeyStore"` provider) that generates/holds keys in a container isolated from the app. On most modern devices this is **hardware-backed** (TEE — Trusted Execution Environment) or **StrongBox** (a dedicated secure element). The app references keys by alias and uses them via `Cipher` / `Signature`; **the raw key never leaves the secure hardware**.
- **Non-extractable keys = the core guarantee.** Even a rooted device / memory dump can't recover a hardware-backed Keystore key. (It *can* potentially *use* the key while the app runs, unless the key is auth-bound — see threat model.)
- **Envelope encryption.** Keystore AES keys can't be exported, so the pattern is: keep a Keystore AES-GCM key per secret-class; use it to encrypt the secret; store the resulting IV + ciphertext in ordinary storage (DataStore/file). To read, decrypt with the Keystore key.
- **AES-GCM.** Authenticated encryption (confidentiality + integrity); requires a unique IV per encryption (never reuse an IV with the same key).
- **Auth-bound keys (`setUserAuthenticationRequired`).** A key can require the user to have authenticated (biometric/device credential) within a time window (or per-use) before `Cipher.init` succeeds — tying secret access to user presence (Ch 21). This is what protects secrets even against someone who has the unlocked running app vs a stolen device.
- **StrongBox (`setIsStrongBoxBacked`).** A separate tamper-resistant secure element; stronger than TEE but not on all devices — request with a fallback.
- **Key attestation.** The Keystore can produce a certificate chain attesting a key is hardware-backed (for server-side verification) — used in high-assurance apps.
- **Key invalidation.** `setInvalidatedByBiometricEnrollment(true)` invalidates an auth-bound key when biometrics change (anti-tamper). Removing the lock screen can also invalidate keys. The app must handle `KeyPermanentlyInvalidatedException` by re-authenticating / re-provisioning.

Core insight: **generate a non-extractable, hardware-backed Keystore key and use it to envelope-encrypt secrets stored in ordinary storage; bind the key to user auth for sensitive secrets.** The Keystore moves the *key* out of reach; your code moves the *secret* out of plaintext.

20.4 High-Level Architecture


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Components & ownership

- **`SecureStore`** — the one abstraction: get-or-create a Keystore key, encrypt/ decrypt secrets, store ciphertext, clear.
- **Android Keystore** — holds the non-extractable key (TEE/StrongBox).
- **Ordinary storage** (DataStore/file) — holds IV + ciphertext (safe, since the key is in hardware).
- **BiometricPrompt** (Ch 21) — unlocks auth-bound keys for sensitive secrets.
- **Backup rules** — exclude secret blobs.

Direction: app → `SecureStore` → Keystore key (use) + ciphertext storage.

20.5 Detailed Component Design



Element	Responsibility	Security property
Keystore AES key (alias)	encrypt/decrypt secrets	non-extractable, hardware-backed
`SecureStore`	envelope encrypt + persist	plaintext never stored
IV + ciphertext blob	durable storage	useless without the Keystore key
auth-bound key (optional)	gate sensitive secrets	requires user auth (Ch 21)
backup exclusion	prevent off-device copy	not in cloud/ `adb backup`

`TokenStore` (Ch 19) is implemented on top of `SecureStore` — the refresh token is just a secret.

20.6 End-to-End Data Flow

Store and read a secret



Auth-bound key invalidated (new biometric enrolled)



20.7 Complete Kotlin Implementation

Package: `com.streamcart.core.security`.

20.7.1 Keystore key generation (AES-GCM, hardware-backed, StrongBox fallback)

```
object KeystoreKeys {
    private const val PROVIDER = "AndroidKeyStore"
    const val TOKEN_KEY = "streamcart.token.key"

    fun getOrCreate(alias: String, requireUserAuth: Boolean = false): SecretKey {
        val ks = KeyStore.getInstance(PROVIDER).apply { load(null) }
        (ks.getEntry(alias, null) as? KeyStore.SecretKeyEntry)?.let { return it.secretKey }

        val spec = KeyGenParameterSpec.Builder(
            alias, KeyProperties.PURPOSE_ENCRYPT or KeyProperties.PURPOSE_DECRYPT
        )
            .setBlockModes(KeyProperties.BLOCK_MODE_GCM)
            .setEncryptionPaddings(KeyProperties.ENCRYPTION_PADDING_NONE)
            .setKeySize(256)
            .apply {
                if (requireUserAuth) {
                    setUserAuthenticationRequired(true)                // auth-bound (Ch 21)
                    setInvalidatedByBiometricEnrollment(true)          // invalidate on new
biometric
                    if (Build.VERSION.SDK_INT >= 30)
                        setUserAuthenticationParameters(0, KeyProperties.AUTH_BIOMETRIC_STRONG)
                }
                if (Build.VERSION.SDK_INT >= 28) setIsStrongBoxBacked(true)   // try StrongBox
            }
            .build()

        return try {
            generate(spec)
        } catch (e: StrongBoxUnavailableException) {               // fallback: TEE
            generate(KeyGenParameterSpec.Builder(alias, /* same, */ spec.purposes)
                .setBlockModes(KeyProperties.BLOCK_MODE_GCM)
                .setEncryptionPaddings(KeyProperties.ENCRYPTION_PADDING_NONE)
                .setKeySize(256).build())
        }
    }

    private fun generate(spec: KeyGenParameterSpec): SecretKey =
        KeyGenerator.getInstance(KeyProperties.KEY_ALGORITHM_AES, PROVIDER)
            .apply { init(spec) }.generateKey()
}
```

20.7.2 Envelope encryption (AES-GCM, unique IV)

```
class Crypto @Inject constructor() {
    private val transform = "AES/GCM/NoPadding"

    fun encrypt(key: SecretKey, plaintext: ByteArray): ByteArray {
        val cipher = Cipher.getInstance(transform).apply { init(Cipher.ENCRYPT_MODE, key) }
        val iv = cipher.iv                                          // GCM: random IV per encryption
        val ct = cipher.doFinal(plaintext)
        return iv + ct                                             // prepend 12-byte IV
    }

    fun decrypt(key: SecretKey, blob: ByteArray): ByteArray {
        val iv = blob.copyOfRange(0, 12)
        val ct = blob.copyOfRange(12, blob.size)
        val cipher = Cipher.getInstance(transform).apply {
            init(Cipher.DECRYPT_MODE, key, GCMParameterSpec(128, iv))   // may throw if auth-
bound (Ch 21)
        }
        return cipher.doFinal(ct)                                  // throws AEADBadTagException if
tampered
    }
}
```

20.7.3 SecureStore (envelope encrypt → DataStore)

```
class KeystoreSecureStore @Inject constructor(
    private val crypto: Crypto,
    private val store: DataStore<Preferences>,     // holds IV+ciphertext (safe)
) : SecureStore {

    override suspend fun put(name: String, secret: String) {
        val key = KeystoreKeys.getOrCreate(KeystoreKeys.TOKEN_KEY)
        val blob = crypto.encrypt(key, secret.toByteArray())
        store.edit { it[stringPreferencesKey(name)] = Base64.encodeToString(blob,
Base64.NO_WRAP) }
    }

    override suspend fun get(name: String): String? {
        val b64 = store.data.first()[stringPreferencesKey(name)] ?: return null
        val key = KeystoreKeys.getOrCreate(KeystoreKeys.TOKEN_KEY)
        return try {
            String(crypto.decrypt(key, Base64.decode(b64, Base64.NO_WRAP)))
        } catch (e: KeyPermanentlyInvalidatedException) {      // biometric changed / lock
removed
            remove(name); deleteKey(); null                    // force re-provision (Ch 19/22)
        } catch (e: AEADBadTagException) { remove(name); null } // tampered/corrupt
    }

    override suspend fun remove(name: String) = store.edit {
it.remove(stringPreferencesKey(name)) }
    override suspend fun clear() {                             // logout (Ch 22) / account
switch (Ch 42)
        store.edit { it.clear() }; deleteKey()
    }
    private fun deleteKey() =
        KeyStore.getInstance("AndroidKeyStore").apply { load(null)
}.deleteEntry(KeystoreKeys.TOKEN_KEY)
}
```

20.7.4 TokenStore (Ch 19) on top of SecureStore

```
class SecureTokenStore @Inject constructor(private val secure: SecureStore) : TokenStore {
    override suspend fun current(): Tokens? {
        val access = secure.get("access") ?: return null
        val refresh = secure.get("refresh") ?: return null
        return Tokens(access, refresh, /* expiresAt */ 0, /* version */ 0)
    }
    override suspend fun update(access: String, refresh: String, expiresAt: Long) {
        secure.put("access", access); secure.put("refresh", refresh)
    }
    override suspend fun clear() = secure.clear()
}
```

20.7.5 Backup exclusion (res/xml)

```
<!-- android:dataExtractionRules (API 31+) / android:fullBackupContent -->
<data-extraction-rules>
  <cloud-backup><exclude domain="file" path="datastore/secure.preferences_pb"/></cloud-backup>
  <device-transfer><exclude domain="file" path="datastore/secure.preferences_pb"/></device-
transfer>
</data-extraction-rules>
```

20.7.6 Module structure

```
:core:security  KeystoreKeys, Crypto, SecureStore(KeystoreSecureStore)
:data:auth      SecureTokenStore (implements TokenStore, Ch 19)
res/xml         data_extraction_rules.xml (backup exclusion)
```

20.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Concurrent encrypt with same key	IV reuse (catastrophic for GCM)	GCM generates a fresh random IV per `Cipher.init`; never hand-set a reused IV
Concurrent get-or-create key	two key generations	Keystore `getEntry` is idempotent by alias; guard with a mutex if needed
Read during key deletion (logout)	decrypt of orphaned blob	`clear()` removes blob + key together; handle missing key gracefully
Multi-process access	inconsistent state	DataStore single-process; keep secret access in one process
Auth-bound decrypt races prompt	`UserNotAuthenticated`	serialize sensitive decrypts behind the BiometricPrompt (Ch 21)

- **IV uniqueness is a correctness/security invariant:** reusing an IV with the same GCM key breaks confidentiality. Always use the cipher-generated IV; store it with the ciphertext; never reuse.

20.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
Biometric re-enrolled (auth-bound key)	`KeyPermanentlyInvalidatedException`	delete key+secret; force re-login/re-provision	re-auth
Lock screen removed	key invalidated	same as above	re-auth
Ciphertext tampered/corrupt	`AEADBadTagException`	discard; re-fetch/re-login	re-auth
StrongBox unavailable	`StrongBoxUnavailableException`	fall back to TEE	invisible
Keystore unavailable (rare OEM bug)	exception on init	degrade (in-memory session only) / error	sign-in required
Secret missing after update	app update/key loss	treat as logged out	re-login
Rooted device	—	Keystore still protects key (hardware); add attestation/root detection (Ch 26) for high assurance	depends



> **Key invalidation is normal, not exceptional:** users change fingerprints and remove PINs. Treat `KeyPermanentlyInvalidatedException` as "re-authenticate," never as a crash — wipe and recover gracefully.

20.10 Scaling the Implementation

- **Level 1.** One Keystore AES key + envelope-encrypt the refresh token into DataStore; exclude from backup.
- **Level 2.** A `SecureStore` abstraction for all secrets; clear on logout.
- **Level 3 — production.** StrongBox with TEE fallback; invalidation handling; per-account scoping (Ch 42); auth-bound keys for sensitive secrets (Ch 21).
- **Level 4 — large.** Key attestation (server verifies hardware-backing); key rotation policy; separate keys per secret-class; DB/media key wrapping (Ch 24).
- **Level 5 — enterprise.** Device-bound tokens (DPoP/mTLS, Ch 19), remote wipe, anomaly/root detection (Ch 26), compliance (PCI/health), HSM-backed server side.

Avoid overengineering: most apps need one Keystore key + envelope encryption + backup exclusion. Attestation, StrongBox, and auth-bound keys are for high-value secrets (payments, health).

20.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Secret storage	Keystore + envelope encryption (manual)	`EncryptedSharedPreferences` (Jetpack Security)	Jetpack Security is **deprecated** — prefer Keystore-direct or Tink
Crypto lib	Keystore + `Cipher` (AES-GCM)	Google **Tink**	Tink for higher-level, misuse-resistant crypto; raw Cipher for control
Key backing	StrongBox	TEE	StrongBox stronger but not universal — request with fallback
Auth binding	`setUserAuthenticationRequired`	unbound	Auth-bound for sensitive secrets (Ch 21); unbound for convenience (silent refresh)
Non-secret config	DataStore plaintext (Ch 4)	SecureStore	Don't encrypt non-secrets; reserve crypto for secrets
Assurance	Key attestation	none	Attestation for high-value/anti-fraud

EncryptedSharedPreferences is deprecated (androidx.security:security-crypto, as of 2024): it's convenient but no longer recommended. Build on the **Keystore directly** (envelope encryption, shown here) or use **Tink** (Google's misuse-resistant crypto library) which also uses the Keystore to protect its master key. Don't reach for the deprecated API in new code.

Auth-bound vs unbound keys: auth-bound keys require user presence to *use* the key (protecting against a stolen-but-unlocked device targeting a specific secret), but break **silent** background refresh (Ch 19) — you can't decrypt the refresh token without a prompt. The common split: refresh token with an **unbound** key (silent refresh), high-value secrets (payment token) with an **auth-bound** key (biometric gate).

20.12 Security Considerations

- **Threat model clarity.** Keystore keys are **non-extractable** (hardware). On a **rooted device with the app unlocked and running**, an attacker may still get the app to *use* an unbound key (decrypt the secret) — this is why sensitive secrets use **auth-bound** keys (require a fresh biometric) and why root detection/attestation (Ch 26) matters for high assurance. Be precise about what the hardware does and doesn't stop.
- **Never hardcode keys** in the APK; never derive a key from a static secret.
- **Exclude secrets from backup** (`dataExtractionRules` / `fullBackupContent`) so they don't leave the device.
- **AES-GCM with unique IVs**; authenticated encryption detects tampering (`AEADBadTagException`).
- **Clear on logout and scope per account** (Ch 22/42) — a lingering valid token is an exposure.
- **No secrets in logs/crash reports** (Ch 28).
- Client vs backend: the Keystore protects secrets **on the device**; the backend must still assume a token can leak (short lifetimes, rotation, revocation — Ch 19). Device storage reduces risk; it doesn't eliminate the need for server-side defenses.

20.13 Performance and Resource Optimization

- **Crypto is cheap** (AES-GCM is fast, often hardware-accelerated); but **cache the `SecretKey` reference** (get-or-create once) rather than reloading per operation.
- **Avoid per-request decrypt** of the token on hot paths — decrypt once, keep in memory for the session (balance against sensitivity).
- **StrongBox is slower** than TEE for key ops; fine for infrequent secret access, avoid for high-frequency operations.
- **Auth-bound keys add a prompt** — don't put them on hot paths (silent refresh).
- **Profiling:** measure key-gen (one-time) and encrypt/decrypt latency; watch for accidental per-call key generation.

20.14 Testing Strategy

Keystore is hard to unit-test on JVM (needs a device/emulator). Abstract behind `SecureStore` and test (a) the contract with a fake, (b) crypto round-trips on device (instrumentation), (c) invalidation handling.

```
// JVM: test consumers against a fake SecureStore (Keystore not available off-device).
class SecureTokenStoreTest {
    private val secure = FakeSecureStore()
    private val store = SecureTokenStore(secure)
    @Test fun `stores and reads tokens`() = runTest {
        store.update("a", "r", 0)
        assertEquals("r", store.current()!!.refresh)
    }
    @Test fun `clear wipes secrets`() = runTest {
        store.update("a", "r", 0); store.clear()
        assertNull(store.current())
    }
}

// Instrumentation (device): real Keystore round-trip + invalidation.
@RunWith(AndroidJUnit4::class)
class KeystoreCryptoTest {
    @Test fun encrypt_decrypt_roundtrip() {
        val key = KeystoreKeys.getOrCreate("test.key")
        val crypto = Crypto()
        val blob = crypto.encrypt(key, "secret".toByteArray())
        assertEquals("secret", String(crypto.decrypt(key, blob)))
    }
    @Test fun tampered_ciphertext_throws() {
        val key = KeystoreKeys.getOrCreate("test.key"); val c = Crypto()
        val blob = c.encrypt(key, "x".toByteArray()).also { it[it.size - 1]++ }  // tamper
        assertThrows(AEADBadTagException::class.java) { c.decrypt(key, blob) }
    }
}
```

What to test on device: hardware-backing (`KeyInfo.isInsideSecureHardware`), StrongBox fallback, `KeyPermanentlyInvalidatedException` recovery (hard to simulate — document manual test), backup exclusion.

20.15 Production Observability

- **Metrics:** key-invalidation rate (biometric re-enroll / lock removed), decrypt- failure rate (`AEADBadTag`), StrongBox-availability rate, secret-access latency, forced-reauth-from-invalidation rate.
- **Logs:** `secure_store{op, result}` — **never** secret values or key material; `key_invalidated{reason}` , `crypto_error{type}` .
- **Debugging:** sudden re-login spikes → key invalidation (biometric/lock change) not handled gracefully; "can't decrypt token" → key deleted/rotated without wiping blob; secrets in a cloud backup → missing backup exclusion.
- **Alerting:** decrypt-failure spike could indicate tampering or a device/OS issue.
- **Privacy:** secrets and key aliases tied to users are sensitive — minimal, redacted logging (Ch 28).

20.16 System Design Interview Preparation

> 40+ questions.

Fundamentals

1. *What is the Android Keystore?* A system key container that generates/holds keys in hardware-isolated storage; keys are non-extractable.
2. *Why not store a token in SharedPreferences?* Plaintext; readable on rooted devices / backups → account takeover.
3. *Why can't you store a raw AES key in the app?* Decompiling/memory-dumping recovers it; "encryption" with a recoverable key is none.
4. *Envelope encryption?* A Keystore key (non-exportable) encrypts the secret; store IV+ciphertext in ordinary storage.
5. *Why AES-GCM?* Authenticated encryption — confidentiality + tamper detection; needs a unique IV.

Keystore internals 6. *TEE vs StrongBox?* TEE = trusted execution environment (common); StrongBox = a dedicated secure element (stronger, not universal) — request with fallback. 7. *What does "non-extractable" guarantee?* The app can *use* the key but can't *read* it, even when rooted. 8. *What does it NOT guarantee?* On a rooted, unlocked, running app, an unbound key can still be *used* to decrypt — hence auth-bound keys + attestation for high value. 9. *Key attestation?* A cert chain proving a key is hardware-backed, verifiable server-side.

Auth-bound keys 10. `setUserAuthenticationRequired`*?* `Cipher.init` requires recent user auth (biometric/credential) → ties secret use to presence (Ch 21). 11. *Per-use vs time-bound auth?* Per-use requires a prompt each time; time-bound allows a window. 12. `setInvalidatedByBiometricEnrollment`*?* Invalidates the key when biometrics change (anti-tamper). 13. *Trade-off of auth-bound keys?* Secure but breaks silent refresh — can't decrypt without a prompt.

Pattern 14. *Where do IV + ciphertext go?* Ordinary storage (DataStore/file) — safe because the key is in hardware. 15. *IV reuse danger?* Reusing an IV with the same GCM key breaks confidentiality; always use a fresh IV. 16. *Store refresh token how?* Unbound Keystore key (silent refresh) + backup exclusion. 17. *Store payment token how?* Auth-bound key (biometric gate, Ch 21).

Invalidation/failure 18. `KeyPermanentlyInvalidatedException`*?* Key invalidated (biometric re-enroll/ lock removed); wipe + re-provision. 19. *How recover?* Delete key+secret; force re-login; don't crash. 20. `AEADBadTagException`*?* Tampered/corrupt ciphertext; discard + re-fetch. 21. *StrongBox unavailable?* `StrongBoxUnavailableException` → fall back to TEE.

Backup/scope 22. *Why exclude from backup?* Secrets shouldn't go to cloud/`adb backup`. 23. *Per-account?* Scope keys/blobs per account; clear on switch (Ch 42). 24. *Logout?* Delete key + ciphertext (and revoke server-side, Ch 19/22).

Deprecation 25. *EncryptedSharedPreferences?* Deprecated (Jetpack Security Crypto) — use Keystore-direct or Tink. 26. *Tink?* Google's misuse-resistant crypto lib; uses Keystore for its master key.

Security model 27. *Does Keystore defeat a rooted device?* It protects the *key material*; it doesn't stop a rooted app from *using* an unbound key while running → add auth-binding/attestation/root detection (Ch 26). 28. *Hardcoded keys?* Never — decompilable. 29. *Client vs backend responsibility?* Device storage reduces leak risk; backend still needs short tokens/rotation/revocation.

Performance 30. *Is crypto a bottleneck?* No (hardware-accelerated); but cache the key reference, don't regenerate per op. 31. *StrongBox perf?* Slower key ops; fine for infrequent access.

Testing 32. *Why hard to unit-test?* Keystore needs a device/emulator; abstract behind `SecureStore` and test consumers with a fake. 33. *How test crypto?* Instrumentation round-trip + tamper → `AEADBadTag`. 34. *How test invalidation?* Document manual (re-enroll biometric); assert wipe path.

Production debugging 35. *Re-login spikes?* Unhandled key invalidation. 36. *Can't decrypt token?* Key rotated/deleted without wiping blob. 37. *Secrets in backup?* Missing exclusion rules.

Design 38. *Which keys auth-bound?* High-value (payments); not the refresh token. 39. *Rotate keys?* Re-encrypt secrets under a new key; handle old-key decrypt during migration. 40. *Attestation when?* High-assurance/anti-fraud flows.

Design-change challenges

1. **Payment token must require biometric to use.** Auth-bound Keystore key + BiometricPrompt (Ch 21).
2. **Encrypt the offline DB.** Wrap the SQLCipher key with a Keystore key (Ch 24).
3. **Detect compromised devices.** Key attestation + root detection (Ch 26).
4. **Rotate the storage key.** Re-encrypt under a new alias; migrate; delete old.
5. **Multi-account secrets.** Per-account aliases/blobs; clear on switch (Ch 42).

Mock walkthrough (abridged)

Clarify (what secrets, threat model, silent vs gated) → Keystore key (non-extractable, StrongBox/TEE) → envelope-encrypt into ordinary storage → auth-bind sensitive secrets (Ch 21) → backup exclusion + per-account + clear on logout → invalidation handling → be precise on what hardware does/doesn't protect → testing.

20.17 Practical Exercises

1. **Beginner.** Generate a Keystore AES-GCM key and envelope-encrypt a token into DataStore; exclude it from backup. *Eval:* no plaintext; key non-extractable (`isInsideSecureHardware`).
2. **Intermediate.** Build a `SecureStore` + `SecureTokenStore` (Ch 19); clear on logout (key + blob). *Eval:* logout wipes everything.
3. **Advanced.** Add an auth-bound key for a "payment token" requiring biometric (Ch 21) and handle `KeyPermanentlyInvalidatedException`. *Eval:* gated access; graceful invalidation recovery.
4. **Production debugging.** Fix an app storing a refresh token in plaintext prefs and backing it up to the cloud. *Eval:* encrypted + excluded from backup.
5. **System design challenge.** Rotate the storage key without losing data (re-encrypt under a new alias). *Eval:* secrets survive rotation; old key deleted.

(Solutions withheld unless requested.)

20.18 Chapter Summary & Reference Sheet

Key decisions: store secrets via **envelope encryption** with a **non-extractable Android Keystore** AES-GCM key (StrongBox with TEE fallback); keep **IV + ciphertext in ordinary storage**; **auth-bind** keys for high-value secrets (Ch 21) but keep the refresh token on an **unbound** key for silent refresh; **exclude secrets from backup**; **clear on logout** and **scope per account**; handle **key invalidation** (`KeyPermanentlyInvalidatedException`) gracefully; **never hardcode keys or log secrets**; **don't use the deprecated** `EncryptedSharedPreferences` — Keystore-direct or Tink.

Concepts: Android Keystore, TEE/StrongBox, non-extractable keys, envelope encryption, AES-GCM + unique IV, auth-bound keys, key attestation, key invalidation, backup exclusion.

Common mistakes: plaintext secrets; hardcoded/derived-from-static keys; IV reuse; auth-binding the refresh token (breaks silent refresh); not excluding from backup; not clearing on logout; crashing on key invalidation; using the deprecated EncryptedSharedPreferences; over-claiming what the Keystore protects on a rooted device.

Checklist: ☐ Keystore AES-GCM key ☐ StrongBox + TEE fallback ☐ envelope encrypt (IV+ciphertext) ☐ unique IV ☐ auth-bind sensitive secrets ☐ backup exclusion ☐ clear on logout + per-account ☐ handle `KeyPermanentlyInvalidated` ☐ no hardcoded keys ☐ no secret logging.

Key APIs: `KeyStore("AndroidKeyStore")`, `KeyGenParameterSpec`, `setIsStrongBoxBacked`, `setUserAuthenticationRequired` / `Parameters`, `setInvalidatedByBiometricEnrollment`, `KeyGenerator`, `Cipher("AES/GCM/ NoPadding")`, `GCMParameterSpec`, `KeyPermanentlyInvalidatedException`, `AEADBadTagException`, `KeyInfo.isInsideSecureHardware`, `dataExtractionRules`.

References:

- Android Keystore: https://developer.android.com/privacy-and-security/keystore
- KeyGenParameterSpec: https://developer.android.com/reference/android/security/keystore/KeyGenParameterSpec
- Key attestation: https://developer.android.com/privacy-and-security/security-key-attestation
- Jetpack Security (deprecation note): https://developer.android.com/jetpack/androidx/releases/security
- Tink: https://developers.google.com/tink

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; Keystore key-gen + envelope crypto + SecureStore + TokenStore + backup rules; concurrency (IV uniqueness); invalidation failure matrix; 5 levels; Keystore-vs- deprecated-API and auth-bound-vs-unbound trade-offs; precise threat model; perf (cache key); device + JVM tests; observability; 40 Q&A + 5 challenges. ✅
- **Expand later:** biometric prompt flow (Ch 21), DB/media encryption (Ch 24), root detection/attestation (Ch 26), key rotation.
- **Interview coverage:** exceeds ≥40 bar. **Defines the `SecureStore` / `TokenStore` contract used by Ch 19/22/42.**